Alexa VanHattum

avh@cs.cornell.edu | cs.cornell.edu/~avh

Research Interests

Programming languages, compilers, formal methods

Education

Fall 2018–Present **Cornell University** | Ph.D. Candidate, Computer Science
Estimated graduation: July 2023

Fall 2021 **Cornell University** | M.S., Computer Science | GPA: 4.00

2012–2016 **Brown University** | B.Sc., Computer Science | GPA: 3.80

Publications

ICSE SEIP 2022 *Verifying Dynamic Trait Objects in Rust*
Alexa VanHattum, Daniel Schwartz-Narbonne, Nathan Chong, and Adrian Sampson. International Conference on Software Engineering, Software Engineering in Practice track (ICSE SEIP 2022). PDF.

ASPLOS 2021 *Vectorization for Digital Signal Processors via Equality Saturation.*
Alexa VanHattum, Rachit Nigam, Vincent T. Lee, James Bornholt, Adrian Sampson. International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS 2021). PDF. DOI.

LCTES 2020 *A Synthesis-aided Compiler for DSP Architectures (WiP Paper).*
Alexa VanHattum*, Rachit Nigam*, Vincent T. Lee, James Bornholt, Adrian Sampson (*equal contribution). Conference on Languages, Compilers, and Tools for Embedded Systems (LCTES 2020). PDF. DOI.

Awards & Grants

April 2020–Present **NSF Graduate Research Fellow 2020**

April 2020 **Finalist: Qualcomm Innovation Fellowship 2020** with Rachit Nigam

May 2019 **Outstanding Teaching Assistant** from Cornell CS, for Programming Languages & Logics

April 2019 **Travel Scholarship** CRA-W Grad Cohort for Women Workshop

September 2018 **Travel Scholarship** PLMW at ICFP

May 2016 **Honorable Mention: Senior Prize in CS** Brown CS

October 2015 **Anita Borg Scholar** Grace Hopper Celebration of Women in Computing

Research Positions

Fall 2018–Present **Cornell University**, Graduate Research

	Advisor: Adrian Sampson Creating solver-aided compilers for heterogeneous hardware targets.
May–Dec 2021	**Amazon Web Services**, Automated Reasoning Applied Scientist Intern Mentors: Daniel Schwartz-Narbonne, Nathan Chong Improving dynamic dispatch for the Kani Rust Verifier.
Spring 2016	**Brown University**, Independent Study in Programming Languages Advisor: Shriram Krishnamurthi (Abstract) Used refinement types in Liquid Haskell to matrix arithmetic in R.
Summer 2014	**The University of Toledo**, Physics NSF REU Intern Advisor: Jaques G. Amar (Abstract) Implemented kinetic Monte Carlo simulations of particle island nucleation.

Prior Industry Experience

2016–2018	**Apple**, Health Software Engineer Lead app/database engineer for Apple's first FDA-regulated irregular heart rhythm notification feature. Presented at Apple's WWDC 2017.
Summer 2015	**Microsoft**, Software Engineering Intern Created a message search feature for the Windows Messaging Application.

Teaching Experience

Summer 2022	**Object-Oriented Programming and Data Structures**, Cornell University, Instructor of Record, estimated 90 students
Fall 2020	**Programming Languages & Logics**, Cornell University, Graduate TA
Fall 2018	**Programming Languages & Logics**, Cornell University, Graduate TA
Spring 2016	**Logic for Systems**, Brown University, Head TA
Fall 2017	**Intro. Object-Oriented Programming**, Brown University, Head TA
Spring 2015	**Discrete Structures & Probability**, Brown University, TA
Fall 2014	**Intro. Object-Oriented Programming**, Brown University, TA
Spring 2014	**First Byte of Computer Science**, Brown University, TA

Invited Talks

April 2022	**Verifying Dynamic Trait Objects in Rust** Brown University
April 2022	**Verifying Dynamic Trait Objects in Rust** University of Texas at Austin
December 2021	**Vectorization for DSPs via Equality Saturation** Wellesley College
April 2021	**Vectorization for DSPs via Equality Saturation** Brown University
April 2021	**Guest Lectures: Finite State Machines, Graph Theory** Contra Costa Community College

December 2020	**Vectorization for DSPs via Equality Saturation** University of California Santa Cruz

Service & Outreach

Aug 2019–Present	**Graduate Students for Gender Inclusion in Computing** Executive board (current Vice President)
Mar–June 2022	**EGRAPHS Workshop at PLDI 2022** Program Committee (invited)
Summer 2020	**SPLASH OOPSLA 2020** Artifact Evaluation Committee
Aug 2020–Dec 2021	**Programming Languages Discussion Group** Co-organizer
Jan 2019–Dec 2021	**Cornell CS Colloquium** Student host organizer
Aug-Dec 2020	**CS Teaching Committee** Student representative
June 2019	**PLDI 2019** Student volunteer
Spring 2019	**Cornell CS PhD Visit Day** Co-organizer
Spring 2019	**Cornell Girls' Adventures in Math** Student volunteer
2016–2017	**Apple Women in Science and Engineering Mentoring** Mentor